

CSCI 3110 Term Project Fall 2026

Term Project

Overview

For your term project, you will work in teams of 3 students to design and develop a comprehensive ASP.NET Core MVC Web Application. This project will allow you to apply and integrate various web development concepts and technologies covered in the course.

Team Formation

- Teams must consist of 3 students.
- Students may form their own teams or request to be assigned to a team by the instructor.
- Team formation must be completed by the start of week 3.
- Team members may receive a different grade for all deliverables.

Learning Objectives

By completing this project, you will:

1. Design and implement a complex data model.
2. Develop a full-stack web application using ASP.NET Core MVC.
3. Create and consume a RESTful Web API.
4. Implement client-side interactivity using JavaScript and AJAX.
5. Apply modern front-end frameworks to create responsive and accessible user interfaces.
6. Practice software development best practices, including version control and documentation.

Milestones and Deliverables

Project Proposal (10 points)

- One-page document describing your application idea.
- Preliminary UML class diagram of your data model.
- List of main functionalities as user stories.
- List of technologies you plan to use.
- AI Disclosure.
- **Due: Week 4**
- **Note:** One document submitted per team.

User Stories (20 points)

- User stories and detailed story flows covering all major functionalities.
- AI Disclosure.
- **Due: Week 8**
- **Note:** One document submitted per team.

The Completed Application (70 points)

- Submit your GitHub clone link.
- AI Disclosure.
- **Due: Week 15**
- **Note:** One submission per team.

Final Presentation (Required, graded as part of application)

- Teams will present their applications to the class during Week 15.
- Presentation format: 20-30 minutes per team (slides + live demonstration).
- The live demonstration will be evaluated as part of the **Application Functionalities** grade.
- Both team members must participate in the presentation.

Evaluation Criteria

Your completed application will be evaluated based on the following:

Data Model (5%)

- Design a database schema with at least 3 interconnected entities.
- Implement at least one many-to-many relationship.
- Use Entity Framework Core for database operations.
- Implement data validation using data annotations and server-side validations.

Application Functionalities (50%)

- Create a home page, make it meaningful to your application.
- Implement CRRUD (Create, Read-All, Read, Update, Delete) operations for all entities.
- Develop at least one complex functionality involving multiple entities.
- Implement user authentication and authorization.
- Use your REST API to implement the functionalities.
- Successfully demonstrate all key features during the Week 15 presentation.

REST Web API (15%)

- Design and implement a RESTful API.
- Include proper HTTP method usage (GET, POST, PUT, DELETE).

JavaScript and AJAX (15%)

- Implement at least two features using AJAX and JavaScript for asynchronous operations.

User Interface and UX (10%)

- Use a CSS framework (e.g., Bootstrap, Tailwind CSS) for a responsive design.
- Implement a custom theme or significant visual customizations.
- Ensure your application is accessible (follow WCAG 2.1 AA standards).
- Design and implement an intuitive navigation system.

Documentation and Best Practices (5%)

- Write comprehensive comments for controllers, service classes, and JavaScript code.
 - State the purpose of each class, each method, and functions.
 - Describe each parameter.
 - If your block of code is not self-explanatory, then write short inline comments to explain it.
- Use Git for version control.
 - Create a readme.md document that explains your application, you are to include:
 - Your project proposal
 - Your AI disclosure
 - Accessibility principles followed

Additional Notes

- ✓ This is a team project. All code and design must be the work of your team only. Teams may discuss concepts with other teams, but all implementation must be original.
- ✓ Cite any external sources or libraries used in your code comments.
- ✓ Regular check-ins during office hours with the instructor are encouraged to ensure you're on the right track.
- ✓ Creativity and going beyond the basic requirements are encouraged and may result in maximum points. If you are aiming for an 'A' then you must go beyond the basic requirements.

AI Use

You are permitted to use AI tools for learning and understanding, but not for generating direct solutions to this assignment. The work you submit must be your team's own original creation, demonstrating your team's understanding and application of course materials.

Required Disclosure

You must include a brief statement in your submission describing:

- Which AI tools did you use?
- How did you use them?
- What specific insights or assistance have you gained?
- If you did not use any AI tools, please disclose as such.

Resources

- ✓ ASP.NET Core Documentation: <https://docs.microsoft.com/en-us/aspnet/core/>
- ✓ Entity Framework Core: <https://docs.microsoft.com/en-us/ef/core/>
- ✓ JavaScript MDN Web Docs: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- ✓ WCAG 2.1 Guidelines: <https://www.w3.org/WAI/WCAG21/quickref/>

Good luck with your project! Remember, this is an opportunity to create something impressive for your job application portfolio.